

Vital Portal: A Symptom-Based Disease Prediction System with Contextual Care Guidance

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Abstract— Today, when people experience health symptoms, they usually search on the internet instead of visiting a doctor. However, most of the information available online is either confusing, incomplete, or not reliable, which can increase fear, misunderstanding, and unnecessary anxiety among users. This creates a gap between initial symptom detection and proper medical consultation, where users lack clear and structured guidance.

The proposed system, called *Vital Portal*, aims to address this gap by providing an intelligent and user-friendly symptom analysis platform. The system uses machine learning algorithms such as Random Forest and Decision Tree to predict possible diseases based on user-selected symptoms. In addition to prediction, it provides supportive information including precautions, nutrition advice, and general care suggestions. A chatbot module is also integrated to assist users by answering queries and guiding them in a simple conversational manner.

The main findings of this system show that combining prediction with guidance improves user understanding and reduces confusion during early symptom stages. Unlike traditional systems that only provide disease prediction, Vital Portal offers a more complete and supportive experience for users. The system is designed to be lightweight, easy to use, and accessible without requiring medical expertise.

The main contribution of this work is a hybrid approach that integrates machine learning-based prediction with interactive guidance and chatbot support, helping users make better decisions before consulting a doctor.

Keywords — Machine Learning, Disease Prediction, Symptom-Based Analysis, Healthcare System, Chatbot, Random Forest, Decision Tree, Early Diagnosis.

I. Introduction

Nowadays, many people prefer to search online when they experience health problems such as fever, headache, or body pain. Instead of visiting a doctor immediately, they try to understand their symptoms through search engines. However, most of the information available online is either too general or not verified, which can lead to confusion and unnecessary fear. This situation is often called cyberchondria, where people become more anxious after reading medical information online [6].

It has been observed that there is a gap between the time a person notices symptoms and the time they consult a doctor. During this period, people need simple and reliable guidance to understand their condition. Many existing systems focus only on predicting diseases but do not provide complete support such as precautions, nutrition advice, or proper guidance [7], [20].

The proposed system, called Vital Portal, helps users understand their symptoms in a better way. It uses machine learning algorithms such as Random Forest and Decision Tree to predict possible diseases based on user input [1], [8]. In addition to prediction, the system also provides basic guidance such as precautions, nutrition suggestions, and general care tips. A chatbot feature is also included to assist users by answering their queries in a simple manner [18], [19].

The main aim of this system is not to replace doctors but to reduce confusion and help people take better decisions during the early stage of health problems.

II. LITERATURE SURVEY

Many research works have been carried out in the field of disease prediction using machine learning techniques. Most of these systems aim to help users or doctors identify diseases at an early stage based on symptoms or medical data [1], [22]. Different approaches have been used, including basic machine learning models, deep learning methods, chatbot systems, and IoT-based healthcare solutions [2], [21].

Some of the earlier systems are based on simple machine learning algorithms such as Decision Tree, Naïve Bayes, and K-Nearest Neighbors [8], [20]. These systems take symptoms as input and provide the predicted disease as output. It has been observed that these systems are easy to implement and give good accuracy for limited datasets. However, they mostly provide only the disease name and do not guide the user further. There is no clear information about what to do next, which reduces their practical usefulness.

In some research works, more advanced models such as Random Forest and Support Vector Machines are used to improve prediction accuracy [1], [22]. These models can handle larger datasets and identify patterns more effectively. While these systems perform better in terms of prediction, they still focus mainly on accuracy and do not provide additional support such as precautions or health guidance.

Recent studies have also explored deep learning techniques for disease prediction [9], [16], [17]. These methods are capable of handling complex and high-dimensional data, especially in cases like medical imaging and time-series health data. Deep learning models often provide higher accuracy compared to traditional methods. However, they require large datasets, high computational power, and are difficult to implement in real-time applications. These systems are more suitable for research environments rather than for common users.

Another important area of research is ensemble learning, where multiple machine learning models are combined to improve performance [1]. Techniques such as bagging,

boosting, and stacking are used in these systems. Studies have shown that ensemble methods can provide better prediction accuracy compared to single models. However, these systems mainly focus on improving model performance and do not consider user interaction or ease of use.

Some researchers have developed IoT-based healthcare systems where data is collected from sensors such as heart rate monitors, temperature sensors, and wearable devices [2], [21]. This data is then analyzed using machine learning algorithms to predict health conditions. These systems are useful for continuous monitoring and real-time analysis. However, they depend heavily on hardware devices and are not easily accessible to all users. In addition, they require proper setup and maintenance, which makes them less practical for general use.

Chatbot-based healthcare systems have also been developed to improve user interaction [18], [19]. These systems allow users to enter their symptoms in the form of a conversation. The chatbot then responds with possible suggestions or advice. While this improves user experience, many chatbot systems lack strong prediction models and may not always provide accurate results. In some cases, they give only general advice instead of specific guidance.

There are also mobile application-based systems that provide disease prediction features [20]. These applications are easy to use and accessible to many users. However, most of these systems support only a limited number of diseases and symptoms. They often provide only prediction results without detailed explanation or further guidance.

Some research works focus on predicting specific diseases such as heart disease, diabetes, or mental health conditions [3], [5]. These systems are designed for particular datasets and provide good results for those diseases. However, their scope is limited, and they cannot be used as a general solution for multiple health conditions.

In addition, some systems aim to support doctors by providing clinical decision support [2]. These systems analyze patient data and assist in diagnosis. While they are useful in hospitals, they are not designed for direct use by common users. They often require medical knowledge and structured clinical data, which makes them less suitable for the general public.

From these studies, it is clear that significant progress has been made in disease prediction using machine learning. However, most systems focus mainly on prediction accuracy and technical performance. Very few systems provide a complete solution that includes prediction, guidance, and user-friendly interaction.

It has been observed that users need more than just prediction results. They require simple explanations, basic precautions, and guidance on the next steps. There is also a need for systems that are easy to use, accessible without special devices, and supportive during the early stage of symptoms.

Therefore, there is a need for a system that combines machine learning-based prediction with user-friendly features such as guidance, chatbot support, and easy accessibility. This understanding forms the basis for the proposed system, which aims to provide a more complete and practical healthcare solution.

III. RESEARCH GAP

From the literature survey, it has been observed that most existing systems mainly focus on predicting diseases based on symptoms or medical data. While many models achieve good accuracy, they often provide only the predicted disease without offering further guidance to the user.

In many cases, systems do not include important support such as precautions, nutrition advice, or clear next steps. Some approaches depend on complex data like clinical reports or sensor-based inputs, which are not easily available to common users. Other systems focus only on specific diseases, making them limited in scope.

In addition, user interaction is not given much importance. Many systems lack simple interfaces or interactive features such as chatbot support, which can help users understand their condition more easily.

Therefore, there is a need for a system that not only predicts diseases but also provides proper guidance, is easy to use, and supports users during the early stage of symptoms. Such a system should give simple and clear information instead of only prediction results.

IV. CONCLUSION

Several studies have been analysed for disease prediction using machine learning, and their limitations have been identified. Most systems focus mainly on prediction accuracy but do not provide complete support to users. It has been observed that users need simple guidance along with prediction, especially in the early stage of symptoms.

To address this, the proposed system, called Vital Portal, combines disease prediction with additional features such as precautions, nutrition suggestions, and chatbot-based guidance. The system is designed to be simple and easy to use so that users can understand their condition without confusion.

The main aim of this work is to reduce unnecessary fear and help users take better decisions before consulting a doctor. In the future, the system can be improved by increasing the number of diseases, enhancing prediction accuracy, and adding more real-time support features. The system can be useful for common users in understanding their health condition at an early stage.

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